Hyperspectral Anomaly Detection Based on Tensor Approximation With Tensor Double Nuclear Norm

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***Abstract*— In hyperspectral anomaly detection (HAD), tensor low-rankness is essential for effectively separating background and anomaly. However, most of the current low-rank-based methods do not use the spatial–spectral low-rankness and the nonlocal self-similarity simultaneously. To address this issue, we propose a tensor double nuclear norm-based tensor approximation (TDNN-TA) model with all the priors in a unified convex framework, which can be efficiently handled through a well-organized alternating direction method of multipliers. Especially, to thoroughly model the background by tensor approximation, we propose a tensor double nuclear norm (TDNN), which achieves a more precise and flexible exploration of low-rankness and nonlocal self-similarity by applying different low-rank constraints to the global tensor and the group tensor. Moreover, to explore the intrinsic characteristics of different priors by various tensor ranks, we employ the Fourier transform-based three-directional tensor nuclear norm to approximate the nonlocal group tensor rank, and the framelet-based three-modal tensor nuclear norm to approximate the global tensor rank. Experimental results validated on several real hyperspectral datasets demonstrate that TDNN-TA is effective in detecting different sizes of anomalous targets and achieves competitive results for various scenes.**

***Index Terms*— Alternating direction method of multipliers, hyperspectral anomaly detection (HAD), low-rankness, nonlocal self-similarity.**

I. Introduction

HYPERSPECTRAL images (HSIs) capture information across hundreds of contiguous spectral bands, offering detailed spatial and spectral information. They are applied in a diverse range of military and civilian fields [1]. Hyperspectral anomaly detection (HAD) enables the identification of potentially valuable targets amidst complex datasets in HSIs, playing a significant role in HSI applications. Therefore, it is imperative to develop HAD techniques to find targets without prior knowledge.

Over the past few decades, based on the investigation into the statistical characteristics of HSI, the Reed–Xiaoli (RX) algorithm [2], the classical method, has been developed. The RX detector computed the Mahalanobis distance to measure the dissimilarity between testing pixels and their adjacent background [3]. It is a very groundbreaking method, which has inspired various variants such as global RX and local RX [4], dual-window-based RX [5], and multiple-window-based RX [6]. However, the RX algorithm assumes that the background image follows a multidimensional Gaussian statistical model, which is challenged by the complexity and variability of hyperspectral data in real-world applications [7]. To alleviate this problem, a number of improvements to the RX have been proposed. The subspace RX [8] employed principal component analysis (PCA) on the background covariance matrix. The segment RX [9] focused on key factors influencing local background covariance matrix estimation. The weighted RX detector and the linear filter-based RX detector [10] were proposed to estimate the background by altering the weights or filtering the wrong anomalies. The kernel RX [11] introduced a kernelization procedure enabling the mapping of raw data into high-dimensional space. The cluster kernel RX [12] employed a fast eigendecomposition algorithm to classify background pixels into different clusters. Despite the simplicity of RX and RX-based variants, they must satisfy statistical assumptions that limit their applications in more topologically complex scenes [13].

Recently, prior-based methods have been developed, since they do not need the statistical assumptions and realize HAD by exploring the intrinsic properties of HSI. Since the infrequent anomalous targets often hide in locally homogeneous backgrounds, the background exhibits low-rankness across all three dimensions, and the anomaly has a sparse property in spatial dimensions. With these two prominent characteristics in mind, Chen et al. [14] applied the robust PCA framework [15] to decompose HSI data into the low-rank representation (LRR) of the background part and the sparsity of anomalies part. Subsequently, several approaches focusing on low-rank and sparse matrix decomposition (LRaSMD) were proposed [16], [17]. However, these methods often overemphasize the sparse component while neglecting the low-rank background part. To address this issue, Zhang et al. [18] introduced an LRaSMD-based Mahalanobis distance (LSMAD) method to leverage the low-rank prior knowledge of the background. Xu et al. [19] utilized a novel dictionary construction strategy to characterize the background information. Niu and Wang [20], Tan et al. [21], and Cheng and Wang [22] improved the LRR of background by utilizing a learning dictionary incorporating a spatial constraint, or integrating

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the graph regularization and total variation (TV). Xu et al. [23] fused collaborative representation and Euclidean distance into the LRaSMD framework-solving procedure. Li et al. [24] explored LRaSMD under the assumption of a mixture of Gaussian (MoG) and devised a global detector by computing the Manhattan distance. Wang et al. [25] proposed a locally enhanced low-rank prior for HAD model capitalizing on the enhanced low-rank property of background signals in local homogeneous regions. Zhao et al. [26] utilized the Schatten-p norm to approximate both the rank and structured sparse norm, effectively constraining the low-rank characteristics of background and the sparsity of anomalies. These methods have yielded promising outcomes, demonstrating the efficacy and potential of the low-rankness and sparsity in HAD. Yan et al. [27] present a fast and robust HAD algorithm that combines coarse-to-fine band selection and adaptive spatial–spectral feature extraction. However, the methods above typically involve transforming the three-order tensor into a two-order matrix, which is unavoidable to result in the loss of its intrinsic high-order structures of HSI and ignores the nonlocal self-similarity in HSI background.

Consequently, more and more researchers have focused on the three-order tensor-based methods in recent years [28], [29], [30]. The classic Tucker decomposition has caught the attention of researchers. The low-rank tensor decomposition-based anomaly detection model utilized the Tucker decomposition to derive the core tensor of the low-rank component and the unmixing approach to extract the spectral signatures of anomalies [31]. Based on the same Tucker decomposition, Zhang et al. [32] employed a reconstruction-error-based method to eliminate background pixels while retaining anomaly information. Chen et al. [33] introduced a tensor PCA (TPCA)-based method to extract the principal component part and the residual part. Qin et al. [34] proposed a low-rank and sparse tensor decomposition method. However, it is worth noting that Tucker decomposition has an inherent limitation due to its computation. The challenge lies in predefining three ranks before conducting the computation, which can pose practical difficulties [35]. To overcome this problem, tensor LRR is utilized widely. The prior-based tensor approximation (PTA) [36] had successfully incorporated a linear TV norm, a novel truncated nuclear norm, and a $l_{2,1}$ norm. The PCA-tensor low-rank and sparse representation (PCA-TLRSR) [3] employed a weighted tensor nuclear norm, a meticulously crafted dictionary, and a PCA to separate the background and the anomaly. The low average rank with TV regularization model for HAD (LARTVAD) [37] method utilizes tensor singular value decomposition (t-SVD) and tensor nuclear norm to capture the global low-rank structure of the background tensor, while incorporating TV regularization to ensure piecewise smoothness and an $l_{2,1}$-norm to enforce group sparsity on anomalies. The novel HAD-based adaptive low-rank transformed tensor (ALRTT) [38] integrates spatial and spectral information using an adaptive tensor transform, minimizes the nuclear norm on the transformed tensor to preserve the spatial–spectral correlation of the background, and employs $l_{2,1,1}$-norm regularization to effectively depict the group sparsity of anomalies. Nevertheless, methods above using discrete Fourier transform-based tensor nuclear norm also have aspects not considered, that the Fourier transform can only calculate the magnitude of the frequency and lacks the function of locating frequency. The framelet can solve this shortcoming. Moreover, there has been significant progress in numerous tensor-based methods. To address the spatial variability and enhance anomaly detection accuracy, Sun et al. [39] developed the Spatial Invariant Tensor Self-Representation (SITSR) model, which effectively captures both spatial and spectral correlations through a self-representation framework. The generalized nonconvex low-rank tensor representation (GNLTR) model [40] enhances HAD by leveraging a unified nonconvex surrogate function to better approximate the tensor tubal rank and incorporate the $l_{2,1}$-norm for sparse anomaly representation. Moreover, these tensor methods disregard the presence of the nonlocal self-similarity prior. This oversight may lead to limited performance in anomaly detection tasks.

Furthermore, numerous deep learning-based HAD methods maintain the 3-D characteristics inherent in HSI. The low-rank embedded network (LREN) method [41] introduced deep neural networks to establish an unsupervised LREN with a specially crafted latent representation. Meanwhile, the denoising convolutional neural network regularized anomaly detection (DeCNN-AD) method [42] devised a plug-and-play framework by integrating a convolutional neural network denoiser and a novel dictionary derived from clustering techniques. However, the interpretability of these methods based on deep learning is relatively poor, and the exploration of the inherent prior knowledge of HSI is not clear.

In this article, we propose a tensor double nuclear norm (TDNN) model to approximate the low-rankness and the nonlocal self-similarity of HSI. To ensure the low-rankness of HSI, we employ the regularization of the framelet-based three-modal tensor nuclear norm as a constraint to global HSI tensor, which offers a more precise characterization of the tensor low-rankness across the three dimensions compared to traditional tensor nuclear norms. To exploit the nonlocal self-similarity inherent in HSI, we employ three-directional tensor nuclear norm regularization for each group tensor, organized based on patches' similarity, which effectively leverages the nonlocal self-similarity characteristic of HSI while considering computational cost and time constraints. Then, we propose a TDNN-based tensor approximation (TDNN-TA) model for HAD. All priors are integrated into a unified convex framework. The contributions of our work can be summarized as follows.

1) Considering the global low-rankness and nonlocal self-similarity at the same time, we design a TDNN. Tensor low-rank constraints comprehensively and flexibly characterizes the low-rankness and nonlocal self-similarity of HSI, applied to both global tensor and group tensors.
2) Considering the different intrinsic characteristics of different priors, we apply the Fourier transform-based three-directional tensor nuclear norm to approximate the nonlocal HSI tensor rank, which can be accomplished quickly by fast Fourier transform (FFT). Meanwhile, we apply the framelet-based three-modal TNN

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(F-3MTNN) to approximate the global HSI tensor rank, which utilizes the redundancy of the framelet transform, making it more effective in exploiting the low-rankness in three dimensions.

3) Based on TDNN, We propose an efficient TDNN-TA model for HAD, which can be handled by the alternating direction multipliers method (ADMM). Experiments on HSIs demonstrate that the proposed method achieves effectiveness and superiority when compared to state-of-the-art approaches.

The remainder of this article is organized as follows. Notations are introduced in Section II. The proposed TDNN model is presented in Section III. The proposed TDNN-TA and its solving algorithm are presented in Section IV. Section V provides experimental results and discussions. Conclusions are drawn in Section VI.

II. Notations and Preliminaries

A. Notations

In this section, some related operations are generalized.

Generally, we denote the vector as $\mathbf{a} \in \mathbb{R}^n$, the matrix as $\mathbf{A} \in \mathbb{R}^{n_1 \times n_2}$ and the third-order tensor as $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$. The (i, j, t)th element in tensor $\mathcal{A}$ is denoted as $\mathcal{A}(i, j, t)$. $(\mathcal{A})^{(i)}$ is the ith mode-3 slice of $\mathcal{A}$, i.e., $(\mathcal{A})^{(i)} = \mathcal{A}(:, :, i)$. The mode-$k$ permutation of $\mathcal{A}$ is defined as $\mathcal{A}_k = \text{permute}(\mathcal{A}, k)$, $k = 1, 2, 3$, where the mth mode-3 slice of $\mathcal{A}_k$ is the mth mode-p slice of $\mathcal{A}$, i.e., $\mathcal{A}(i, j, t) = \mathcal{A}_1(j, t, i) = \mathcal{A}_2(t, i, j) = \mathcal{A}_3(i, j, t)$. Naturally, its inverse operation is $\mathcal{A} = \text{ipermute}(\mathcal{A}_k, k)$.

Given a tensor $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, the transformed tensor via discrete Fourier transform (DFT) along the kth mode is $\bar{\mathcal{A}}_k$, that is, $\bar{\mathcal{A}}_k = \text{fft}(\mathcal{A}, [], k)$, following the terminology used in MATLAB. Certainly, we can compute $\mathcal{A}$ via $\mathcal{A} = \text{ifft}(\bar{\mathcal{A}}_k, [], k)$.

The unfold and fold operator of tensor $\mathcal{X}$ is defined as $\text{unfold}(\mathcal{X}) = \left[\mathcal{X}^{(1)}, \mathcal{X}^{(2)}, \ldots, \mathcal{X}^{(n_3)}\right]^T$, $\text{fold}(\text{unfold}(\mathcal{X})) = \mathcal{X}$, where the unfold operator can transform tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ into an $n_1 n_3 \times n_2$ matrix along the third dimension, while the fold operator is its inverse operator.

The transformed tensor via framelet transform along the kth mode is $\hat{\mathcal{A}}_k$, that is, $\hat{\mathcal{A}}_k = \text{fold}\left(\mathbf{W_k}(\mathcal{A}_k)_3\right)$, where $\mathbf{W_k} \in \mathbb{R}^{wn_k \times n_k}$ is the framelet transform matrix. According to the unitary extension principle (UEP) property [43], we have $\mathcal{A}_k = \text{fold}\left(\mathbf{W_k^T}\hat{\mathcal{A}}_{k3}\right)$. $\|\cdot\|_F$ is the Frobenius norm, which is defined as $\|\mathcal{A}\|_F := \left(\sum_{i,j,t} |\mathcal{A}(i, j, t)|^2\right)^{1/2}$. $\|\cdot p\|_{2,1}$ is the $l_{2,1}$ norm, which is defined as $\|\mathcal{A}\|_{2,1} := \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} \left(\sum_{t=1}^{n_3} \mathcal{A}(i, j, t)^2\right)^{1/2}$.

B. Tensor Rank

In the past decades, a lot of research has been devoted to defining the tensor rank. Among these definitions, the following two definitions have the ability to accurately characterize tensor rank.

Definition 1 (DFT-Based Tensor Fibered Rank [44]): Given a tensor $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, the fibered rank of $\mathcal{A}$ is a vector, denoted as $R_f(\mathcal{A})$, in which kth element is the number of nonzero fibers of $\bar{\mathcal{S}}_k$, where $\bar{\mathcal{S}}_k$ comes from the t-SVD [45] of $\bar{\mathcal{A}}_k$: $\bar{\mathcal{A}}_k = \bar{\mathcal{U}}_k * \bar{\mathcal{S}}_k * \bar{\mathcal{V}}_k^T$.

Definition 1, based on the Fourier transform, achieves a flexible and simultaneous representation for the correlations along different modes [44]. As directly minimizing $R_f(\mathcal{A})$ is NP-hard, the following three directional tensor nuclear norm (3DTNN) is defined as its convex relaxation.

Definition 2 (3DTNN [44]): The 3DTNN of a tensor $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, denoted as $\|\mathcal{A}\|_{\text{3DTNN}}$, is defined as

$$\|\mathcal{A}\|_{\text{3DTNN}} := \sum_{k=1}^{3} \alpha_k \|\mathcal{A}\|_{\text{TNN}_k} \tag{1}$$

where $\alpha_k \geqslant 0 (k = 1, 2, 3)$, $\sum_{k=1}^{3} \alpha_k = 1$ and $\|\mathcal{A}\|_{\text{TNN}_k}$ is the mode-k tensor nuclear norm (mode-k TNN) of $\mathcal{A}$, defined as

$$\|\mathcal{A}\|_{\text{TNN}_k} := \sum_{i=1}^{n_k} \left\| \left(\bar{\mathcal{A}}_k\right)^{(i)} \right\|_* \tag{2}$$

where $\|\cdot\|_*$ is the matrix nuclear norm.

Definition 1 and its convex relaxation Definition 2 are used in numerous tensor LRR models for various tasks [44]. However, it is more important that the framelet will bring in redundancy, which means the transformed tensor has a lower rank [46].

Definition 3 (Framelet-Based Tensor Fibered Rank [46]): Given a tensor $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, the fibered rank of $\mathcal{A}$ is a vector, denoted as $R_f^F(\mathcal{A})$, whose kth element is the number of nonzero fibers of $\hat{\mathcal{S}}_k$, where $\hat{\mathcal{S}}_k$ satisfies $\hat{\mathcal{A}}_k = \hat{\mathcal{U}}_k * \hat{\mathcal{S}}_k * \hat{\mathcal{V}}_k^T$.

Due to the framelet transform has redundancy, Definition 3 explores the low-rank characteristics of tensors more accurately. Similarly, it is NP-hard to minimize the framelet-based tensor fibered rank. The following F-3MTNN is defined as the convex relaxation of the framelet-based tensor fibered rank.

Definition 4 (F-3MTNN [46]): The F-3MTNN of a tensor $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, denoted as $\|\mathcal{A}\|_{\text{F-3MTNN}}$, is defined as

$$\|\mathcal{A}\|_{\text{F-3MTNN}} := \sum_{k=1}^{3} \beta_k \|\mathcal{A}\|_{\text{TNN}_k}^{\text{F}} \tag{3}$$

where $\beta_k \geqslant 0 (k = 1, 2, 3)$, $\sum_{k=1}^{3} \beta_k = 1$ and $\|\mathcal{A}\|_{\text{TNN}_k}^F$ is the framelet-based mode-k TNN of $\mathcal{A}$, defined as

$$\|\mathcal{A}\|_{\text{TNN}_k}^{\text{F}} := \sum_{i=1}^{wn_k} \|(\hat{\mathcal{A}}_k)^{(i)}\|_* \tag{4}$$

where $\|\cdot\|_*$ is the matrix nuclear norm.

III. Tensor Double Nuclear Norm

A. Problem Formulation

HSI data can be presented as a third-order tensor $\mathcal{Y} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, where $n_1 \times n_2$ is the spatial size, and n_3 is the number of spectral bands. We decompose $\mathcal{Y}$ into backgrounds tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ and anomaly tensor $\mathcal{S} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, which is anchored as $\mathcal{Y} = \mathcal{X} + \mathcal{S}$. This observation model effectively preserves the inherent three-order tensor structure without requiring any tensor matricization operations. We segment the HSI tensor into overlapping patches and then organize them into similarity-based groups $\mathcal{X}^n, n \in [1, N]$.

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Each group can be viewed as a tensor with a small size. The nonlocal self-similarity of the background can be exploited by depicting the low-rankness of each group. Considering the low-rankness of $\mathcal{X}$, the background tensor approximation problem can be described with

$$\min_{\mathcal{X},\mathcal{S}} \sum_{n=1}^{N} \text{Rank}\left(\mathcal{X}^n\right) + t \cdot \text{Rank}(\mathcal{X}) \quad \text{s.t. } \mathcal{Y} = \mathcal{X} + \mathcal{S} \tag{5}$$

where rank$(\cdot)$ represents the rank of the tensor; t is a regularization parameter. Model (5) ingeniously transforms the characterization of nonlocal self-similarity into the depiction of group tensor low-rankness. By imposing low-rank constraints, it simultaneously captures both the low-rankness and nonlocal self-similarity priors of the background tensor.

B. Tensor Double Nuclear Norm

In accordance with the tensor rank definition outlined in Section II-B, $R_f(\mathcal{A})$ adeptly captures the interdependencies among various dimensions of HSI, with computational efficiency. Meanwhile, framelet provides a detailed, multiscale decomposition that is particularly helpful in capturing the fine spectral and spatial features present in HSI data. However, framelet decomposition in HSI requires performing convolutions and transformations at multiple scales and orientations, which amplifies the computational load. $R_f^F(\mathcal{A})$ provides a more refined approximation of the tensor rank through framelet characteristics, albeit at the expense of increased computational overhead. To visualize this difference, we design the following experiments.

We use two datasets, which are the Sandiego dataset and the Segundo dataset to compare the fibered rank after FFT transformation and framelet transformation. Their sizes are $100 \times 100 \times 189$ and $250 \times 300 \times 224$, respectively. Since each transformed slice of tensor has numerous small singular values for real image data [47], we only keep the singular values that are greater than the truncation parameter for discussion. The truncation parameter is set to 0.1 or 0.2. The result is shown in Table I. And the visual comparison is shown in Fig. 1. The result shows that the transformed tensor via framelet has a lower fibered rank than that via the Fourier transformation, which indicates that $R_f^F(\mathcal{A})$ exhibits a greater capacity to delineate the low-rank structure of HSI compared to $R_f(\mathcal{A})$. However, framelets introduce considerable redundant information, thereby leading to extended computational durations compared to FFT. This disparity is particularly conspicuous for Segundo, because of its large image size. To harness the benefits of nonlocal self-similarity of HSI, it becomes imperative to partition the HSI into patches. We further devise a similar experiment targeting patches. We divide the above two datasets into $5 \times 5 \times n_3$ patches (n_3 represents the spectral dimension of the datasets), and then cluster these patches into groups based on high feature similarity. There are ten similar patches within each group. We randomly select one group to form a tensor with the size of $25 \times n_3 \times 10$ and calculate the fibered rank after FFT transformation and framelet transformation. The result is shown in Table II. And

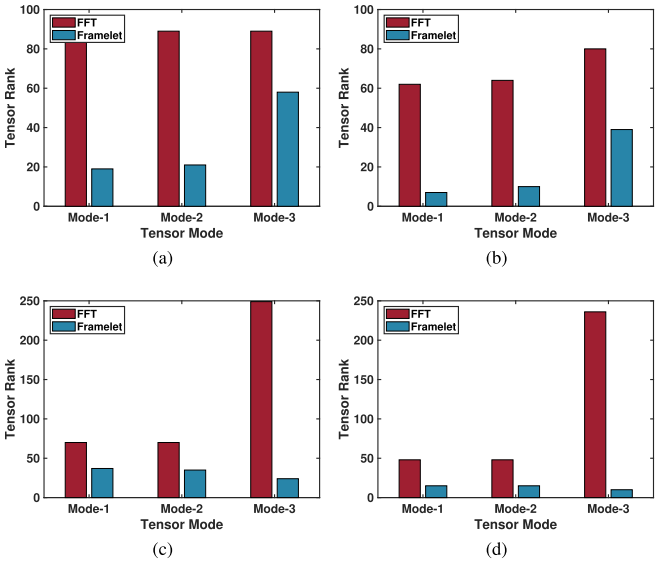


Fig. 1. Fibered rank of HSI tensor for three orders by using a different transform. (a) San Diego ($\tau = 0.1$). (b) San Diego ($\tau = 0.2$). (c) Segundo ($\tau = 0.1$). (d) Segundo ($\tau = 0.2$).

TABLE I
FIBERED RANK OF HSI TENSOR FOR THREE ORDERS BY USING A DIFFERENT TRANSFORM

Data	τ	Transformation	Mode-1	Mode-2	Mode-3	Time(s)
Sandiego	0.1	FFT	87	89	89	0.8769
		Framelet	19	21	58	1.6457
	0.2	FFT	62	64	80	0.8927
		Framelet	7	10	39	1.6510
Segundo	0.1	FFT	70	70	249	8.9688
		Framelet	37	35	24	16.3496
	0.2	FFT	48	48	236	8.9821
		Framelet	15	15	10	16.2167

the visual comparison is shown in Fig. 2. As depicted in Table II, the superiority of the framelet becomes less apparent following a substantial reduction in tensor size. However, the computational time remains lengthier compared to FFT, at times nearly doubling it. As a result, compared to $R_f(\mathcal{A})$, $R_f^F(\mathcal{A})$ more accurately approximates the global low-rank performance of HSI. Nevertheless, it has a limited effect on improving the patches.

Based on the above analysis and the optimization model (5), we propose a TDNN model.

Definition 5 (TDNN): Given a tensor $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, the TDNN $\|\mathcal{A}\|_{\text{TDNN}}$ is denoted as follows:

$$\|\mathcal{X}\|_{\text{TDNN}} = \sum_{n=1}^{N} R_f\left(\mathcal{X}^n\right) + t \cdot R_f^F(\mathcal{X}) \tag{6}$$

where t is the regularization parameters.

In Definition 5, we utilize $R_f^F(\cdot)$ to maintain low-rankness, which has the ability to efficiently leverage the intrinsic low-rank structure of HSI, and flexibly impose constraints across its three dimensions at the same time. To exploit

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TABLE II
FIBERED RANK OF GROUP TENSOR FOR THREE ORDERS BY USING A DIFFERENT TRANSFORM

Data	τ	Transformation	Mode-1	Mode-2	Mode-3	Time(s)
Sandiego Patches	0.1	FFT	9	10	22	0.0873
		Framelet	5	1	11	0.1151
	0.2	FFT	10	10	21	0.0351
		Framelet	5	1	12	0.0737
Segundo Patches	0.1	FFT	8	5	10	0.1994
		Framelet	6	1	11	0.5941
	0.2	FFT	5	2	7	0.1752
		Framelet	5	1	9	0.6791

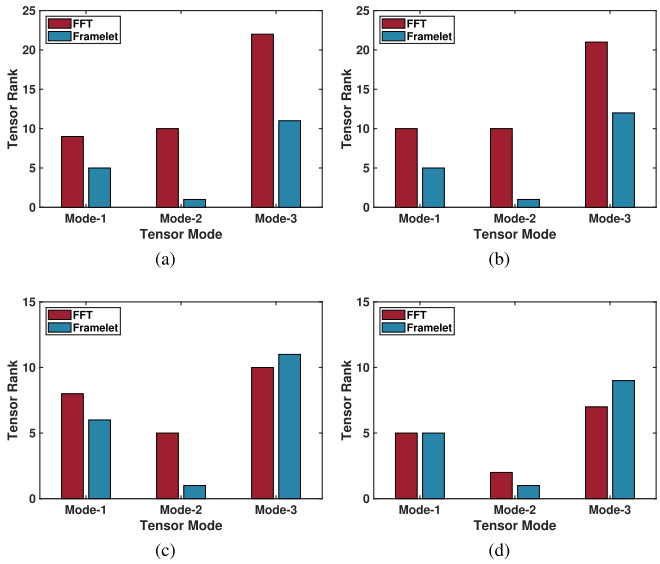


Fig. 2. Fibered rank of group tensor for three orders by using a different transform. (a) San Diego patches ($\tau = 0.1$). (b) San Diego patches ($\tau = 0.2$). (c) Segundo patches ($\tau = 0.1$). (d) Segundo patches ($\tau = 0.2$).

nonlocal self-similarity, we segment the HSI into overlapping patches and then organize them into similarity-based groups. In this step, numerous small patches are extracted from the original HSI, each forming a third-order tensor with a size of $s_p \times s_p \times n_3$, where s_p denotes the patch size and n_p signifies the number of patches in each group. These similar groups are subsequently clustered, with high feature similarity within the same category. Following partitioning and clustering, we can obtain N clustering group tensors with dimensions $s_p^2 \times n_3 \times n_p$. Given the high similarity and small size of each group, we apply $R_f^F(\cdot)$ to each group tensor $\mathcal{X}^n$.

The proposed TDNN applies appropriate tensor rank constraints to the low-rankness of the spatial–spectral dimensions and the nonlocal self-similarity of the tensor. Compared to existing tensor nuclear norm methods, TDNN is more comprehensive and flexible. Specifically, TDNN applies FFT to each clustered tensor group along the spatial, spectral, and self-similarity dimensions, which thoroughly explores the nonlocal self-similarity of the tensor. In addition, TDNN employs a 3-D framelet transformation, enabling it to flexibly and accurately characterize the global low-rankness of the tensor. Compared with the Fourier transform, the framelet can associate frequency intensity information with positional information, thus addressing some of the shortcomings of the Fourier transform. Meanwhile, the framelet is redundant. As a result, the transformed tensor exhibits high similarity, which is beneficial for utilizing low-rank priors.

IV. PROPOSED ANOMALY DETECTION MODEL AND SOLVING ALGORITHM

A. Proposed Anomaly Detection Model

Based on the proposed TDNN, we propose a TDNN-TA model for HAD

$$\min_{\mathcal{X},\mathcal{S}} \sum_{n=1}^{N} R_f\left(\mathcal{X}^n\right) + t \cdot R_f^F(\mathcal{X}) + \lambda\|\mathcal{S}\|_{2,1} \quad \text{s.t. } \mathcal{Y} = \mathcal{X} + \mathcal{S} \tag{7}$$

where t and λ are the regularization parameters.

For the low-rankness and the nonlocal self-similarity, we apply TDNN to the background tensor, which ensures the priors of the background tensor have been effectively explored and balances computational efficiency. In addition, for anomaly sparsity, we enforce constraints using the $l_{2,1}$ norm, ensuring effective sparsity characterization.

We embed the convex relaxation into model (7), the proposed TDNN-TA can be outlined as follows:

$$\min_{\mathcal{X},\mathcal{S}} \sum_{n=1}^{N} \left\|\mathcal{X}^n\right\|_{\text{3DTNN}} + t\|\mathcal{X}\|_{\text{F-3MTNN}} + \lambda\|\mathcal{S}\|_{2,1} \quad \text{s.t. } \mathcal{Y} = \mathcal{X} + \mathcal{S}. \tag{8}$$

Based on Definitions 2 and 4, the proposed model (8) can be represented as

$$\min_{\mathcal{X},\mathcal{S}} \sum_{n=1}^{N}\sum_{k=1}^{3} \alpha_k \left\|\mathcal{X}^n\right\|_{\text{TNN}_k} + t\sum_{k=1}^{3} \beta_k \|\mathcal{X}\|_{\text{TNN}_k}^{\text{F}} + \lambda\|\mathcal{S}\|_{2,1} \quad \text{s.t. } \mathcal{Y} = \mathcal{X} + \mathcal{S} \tag{9}$$

where $\alpha_k \geqslant 0 (k = 1, 2, 3)$, $\sum_{k=1}^{3} \alpha_k = 1$, $\beta_k \geqslant 0 (k = 1, 2, 3)$, $\sum_{k=1}^{3} \beta_k = 1$.

The flow diagram of our TDNN-TA model is illustrated in Fig. 3.

B. ADMM-Based Solving Algorithm

ADMM is a straightforward and efficient approach to solving problem (8), [48], [49]. By introducing the auxiliary variables $\mathcal{Z}_k$, $\mathcal{L}$, (9) is reformulated as

$$\min_{\mathcal{X},\mathcal{S},\mathcal{Z}_k,\mathcal{L},\mathcal{U}} \sum_{n=1}^{N}\sum_{k=1}^{3} \alpha_k \|\mathcal{L}\|_{\text{TNN}_k} + t\sum_{k=1}^{3} \beta_k \|\mathcal{Z}_k\|_{\text{TNN}_k}^{F} + \lambda\|\mathcal{S}\|_{2,1} \quad \text{s.t. } \mathcal{Y} = \mathcal{X} + \mathcal{S}, \quad \mathcal{X} = \mathcal{Z}_k (k = 1, 2, 3), \ \mathcal{L} = \mathcal{X}. \tag{10}$$

The unconstrained augmented Lagrangian function of (10) is converted to the following problem:

$$\min_{\mathcal{X},\mathcal{S},\mathcal{Z}_k,\mathcal{L},\mathcal{U}} \sum_{n=1}^{N}\sum_{k=1}^{3} \alpha_k \|\mathcal{L}\|_{\text{TNN}_k} + t\sum_{k=1}^{3} \beta_k \|\mathcal{Z}_k\|_{\text{TNN}_k}^{F}$$

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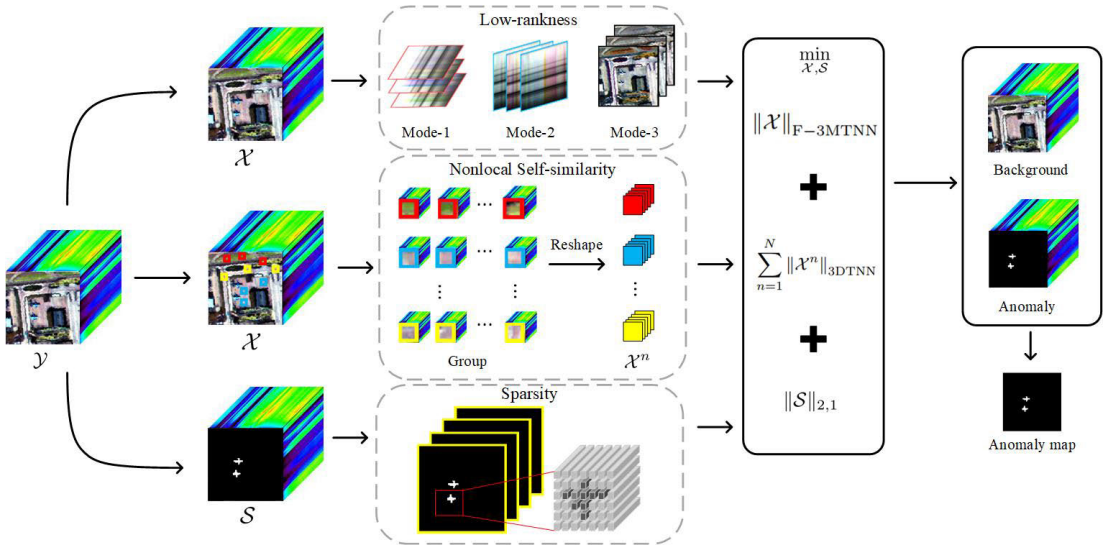


Fig. 3. Flow diagram of our proposed TDNN-TA model for HSI anomaly detection.

$$+\sum_{k=1}^{3}\left\{\frac{\mu_{Lk}}{2}\|\mathcal{L}-\mathcal{X}\|_F^2+\langle M_{Lk},\mathcal{L}-\mathcal{X}\rangle\right\}$$
$$+\sum_{k=1}^{3}\left\{\frac{\mu_{Zk}}{2}\|\mathcal{Z}_k-\mathcal{X}\|_F^2+\langle M_{Zk},\mathcal{Z}_k-\mathcal{X}\rangle\right\}$$
$$+\frac{\mu_Y}{2}\|\mathcal{Y}-\mathcal{X}-\mathcal{S}\|_F^2+\langle M_Y,\mathcal{Y}-\mathcal{X}-\mathcal{S}\rangle$$
$$+\lambda\|\mathcal{S}\|_{2,1} \tag{11}$$

where M_{Lk}, M_{Zk}, and M_Y are the Lagrange multipliers. μ_{Lk}, μ_{Zk}, and μ_Y are the penalty parameters. We solve problem (11) by updating one variable with others fixed.

1) $\mathcal{L}$-Related Subproblem:

$$\begin{aligned}\mathcal{L}&=\underset{\mathcal{L}}{\text{argmin}}\sum_{n=1}^{N}\sum_{k=1}^{3}\alpha_k\|\mathcal{L}\|_{\text{TNN}_k}\\&\quad+\sum_{k=1}^{3}\left\{\frac{\mu_{Lk}}{2}\|\mathcal{L}-\mathcal{X}\|_F^2+\langle M_{Lk},\mathcal{L}-\mathcal{X}\rangle\right\}\\&=\underset{\mathcal{L}}{\text{argmin}}\sum_{n=1}^{N}\sum_{k=1}^{3}\alpha_k\|\mathcal{L}\|_{\text{TNN}_k}\\&\quad+\sum_{k=1}^{3}\frac{\mu_{Lk}}{2}\left\|\mathcal{L}-\left(\mathcal{X}+\frac{M_{Lk}}{\mu_{Lk}}\right)\right\|_F^2.\end{aligned} \tag{12}$$

To solve this problem, we correspondingly segment $\mathcal{X}+(M_{Lk}/\mu_{Lk})$ as $\{\mathcal{X}+(M_{Lk}/\mu_{Lk})\}_{n=1}^{N}$ and obtain the following N problems:

$$\begin{aligned}\mathcal{L}^n&=\underset{\mathcal{L}^n}{\text{argmin}}\sum_{k=1}^{3}\alpha_k\left\|\mathcal{L}^n\right\|_{\text{TNN}_k}\\&\quad+\sum_{k=1}^{3}\frac{\mu_{Lk}}{2}\left\|\mathcal{L}^n-\left(\mathcal{X}^n+\frac{M_{Lk}^n}{\mu_{Lk}}\right)\right\|_F^2\\&=\underset{\mathcal{L}^n}{\text{argmin}}\,\alpha_k\left\|\mathcal{L}^n\right\|_{\text{TNN}_k}+\frac{\mu_{Lk}}{2}\left\|\mathcal{L}^n-\left(\mathcal{X}+\frac{M_{Lk}^n}{\mu_{Lk}}\right)\right\|_F^2.\end{aligned} \tag{13}$$

Based on [44], we introduce the singular value shrinkage operator to solve (13)

$$\mathcal{L}^n=\mathcal{D}^{\text{TNN}}_{\frac{\alpha_k}{\mu_{Lk}}}\left(\mathcal{X}^n+\frac{M_{Lk}^n}{\mu_{Lk}}\right). \tag{14}$$

2) $\mathcal{Z}_k$-Related Subproblem:

$$\begin{aligned}\mathcal{Z}_k&=\underset{\mathcal{Z}_k^n}{\text{argmin}}\;t\sum_{k=1}^{3}\beta_k\left\|\mathcal{Z}_k^n\right\|_{\text{TNN}_k}^{\text{F}}\\&\quad+\sum_{k=1}^{3}\left\{\frac{\mu_{Zk}}{2}\|\mathcal{Z}_k-\mathcal{X}\|_F^2+\langle M_{Zk},\mathcal{Z}_k-\mathcal{X}\rangle\right\}\\&=\underset{\mathcal{Z}_k^n}{\text{argmin}}\;\beta_k\left\|\mathcal{Z}_k^n\right\|_{\text{TNN}_k}^{\text{F}}+\frac{\mu_{Zk}}{2}\left\|\mathcal{Z}_k-\left(\mathcal{X}+\frac{M_{Zk}}{\mu_{Zk}}\right)\right\|_F^2.\end{aligned} \tag{15}$$

To solve (15), we obtain the following results based on the framelet-besed kth-modal singular value shrinkage operator [46] mentioned in our previous article:

$$\mathcal{Z}_k^n=\mathcal{D}^{\text{F-TNN}}_{\frac{\beta_k\cdot t}{\mu_{Zk}}}\left(\mathcal{X}+\frac{M_{Zk}}{\mu_{Zk}}\right). \tag{16}$$

3) $\mathcal{X}$-Related Subproblem:

$$\begin{aligned}\mathcal{X}&=\underset{\mathcal{X}}{\text{argmin}}\sum_{k=1}^{3}\left\{\frac{\mu_{Lk}}{2}\|\mathcal{L}-\mathcal{X}\|_F^2+\langle M_{Lk},\mathcal{L}-\mathcal{X}\rangle\right\}\\&\quad+\sum_{k=1}^{3}\left\{\frac{\mu_{Zk}}{2}\|\mathcal{Z}_k-\mathcal{X}\|_F^2+\langle M_{Zk},\mathcal{Z}_k-\mathcal{X}\rangle\right\}\\&\quad+\frac{\mu_Y}{2}\|\mathcal{Y}-\mathcal{X}-\mathcal{S}\|_F^2+\langle M_Y,\mathcal{Y}-\mathcal{X}-\mathcal{S}\rangle\\&=\underset{\mathcal{X}}{\text{argmin}}\sum_{k=1}^{3}\left\{\frac{\mu_{Lk}}{2}\left\|\mathcal{X}-\left(\mathcal{L}+\frac{M_{Lk}}{\mu_{Lk}}\right)\right\|_F^2\right\}\\&\quad+\sum_{k=1}^{3}\left\{\frac{\mu_{Zk}}{2}\left\|\mathcal{X}-\left(\mathcal{Z}_k+\frac{M_{Zk}}{\mu_{Zk}}\right)\right\|_F^2\right\}\\&\quad+\frac{\mu_y}{2}\left\|\mathcal{X}-\left(\mathcal{Y}-\mathcal{S}+\frac{M_Y}{\mu_Y}\right)\right\|_F^2.\end{aligned} \tag{17}$$

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Obviously, we obtain

$$\mathcal{X} = \left[\sum_{k=1}^{3}\mu_{Lk}\left(\mathcal{L}+\frac{M_{Lk}}{\mu_{Lk}}\right)+\sum_{k=1}^{3}\mu_{Zk}\left(\mathcal{Z}_k+\frac{M_{Zk}}{\mu_{Zk}}\right) +\mu_Y\left(\mathcal{Y}-\mathcal{S}+\frac{M_Y}{\mu_y}\right)\right]\Bigg/\left[\sum_{k=1}^{3}\mu_{Lk}+\sum_{k=1}^{3}\mu_{Zk}+\mu_Y\right]. \tag{18}$$

4) $\mathcal{S}$-Related Subproblem:

$$\begin{aligned}\mathcal{S} &= \underset{\mathcal{S}}{\operatorname{argmin}}\ \lambda\|\mathcal{S}\|_{2,1}+\frac{\mu_y}{2}\|\mathcal{Y}-\mathcal{X}+\mathcal{S}\|_F^2 \\ &\quad +\langle M_Y,\mathcal{Y}-\mathcal{X}-\mathcal{S}\rangle \\ &= \underset{\mathcal{S}}{\operatorname{argmin}}\ \lambda\|\mathcal{S}\|_{2,1}+\frac{\mu_Y}{2}\left\|\mathcal{S}-\left(\mathcal{Y}-\mathcal{X}+\frac{M_Y}{\mu_Y}\right)\right\|_F^2.\end{aligned} \tag{19}$$

The optimization subproblem in (19) can be solved by the $l_{2,1}$ minimization operator according to [50]. Here, we abbreviate its closed-form solution as

$$\mathcal{S}(i,j,:) = Soft_{2,1}\left(\left(\mathcal{Y}-\mathcal{X}+\frac{M_Y}{\mu_Y}\right)(i,j,:),\frac{\lambda}{\mu_Y}\right). \tag{20}$$

5) Update Multipliers: The multipliers M_{Lk}, M_{Zk}, and M_Y are updated as follows:

$$\begin{aligned}M_{Lk} &= M_{Lk}+\tau(\mathcal{L}-\mathcal{X}) \\ M_{Zk} &= M_{Zk}+\tau(\mathcal{Z}_k-\mathcal{X}) \\ M_Y &= M_Y+\tau(\mathcal{Y}-\mathcal{X}-\mathcal{S}).\end{aligned} \tag{21}$$

The convergence condition is

$$\max\left\{\begin{matrix}\left\|\mathcal{Y}-\mathcal{X}^{p+1}-\mathcal{S}^{p+1}\right\|_\infty,\left\|\mathcal{X}^{p+1}-\mathcal{X}^{p}\right\|_\infty \\ \left\|\mathcal{L}^{p+1}-\mathcal{X}^{p+1}\right\|_\infty,\left\|\mathcal{Z}_k^{p+1}-\mathcal{X}^{p+1}\right\|_\infty\end{matrix}\right\}\leqslant\varepsilon. \tag{22}$$

After multiple iterations to reach convergence conditions, the anomaly map can be obtained by

$$\mathbf{T}(i,j)=\sqrt{\sum_{t=1}^{n_3}|\mathcal{S}(i,j,t)|^2}. \tag{23}$$

According to the above-mentioned explanation and derivation, we present the complete optimization procedure for our proposed method in Algorithm 1.

Algorithm 1 TDNN-TA for HSI

Input: Observed HSI $\mathcal{Y}$, $\alpha_k(k=1,2,3)$, $\beta_k(k=1,2,3)$, t, λ, patch size, patch number, ρ, ε, $\mu_{\max}$ and $iter_{\max}$.
Initialize: $\mathcal{L}_k^p=\mathcal{Z}_k^p=\mathcal{X}^p=\mathcal{S}^p=\mathcal{O}$, $M_{Lk}^p=M_{Zk}^p=M_Y^p=\mathcal{O}$, $\rho=1.05$, $\epsilon=1e^{-4}$, $\mu_{\max}=1e^{10}$, $\text{iter}_{\max}=80$.
while not converged and $p\leqslant \text{iter}_{\max}$ **do**
  Update $\mathcal{L}_k^{p+1}$ by (14).
  Update $\mathcal{Z}_k^{p+1}$ by (16).
  Update $\mathcal{X}^{p+1}$ by (18).
  Update $\mathcal{S}^{p+1}$ by (20).
  Update $M_{Lk}^{p+1}=M_{Zk}^{p+1}=M_Y^{p+1}$ by (21).
  Update μ_{Lk}, μ_{Zk} and μ_y:
  $\mu_{Lk}^{p+1}=\rho\mu_{Lk}^p$, $\mu_{Zk}^{p+1}=\rho\mu_{Zk}^p$, $\mu_Y^{p+1}=\rho\mu_Y^p$.
  $p=p+1$.
  Check the convergence (22)
end while
Output: anomalies $\mathcal{S}$ and detection map $\mathbf{T}$.

V. Experiments

In this section, we present several HSIs to evaluate the detection efficacy and performance of our proposed TDNN-TA model. All experimental outcomes were acquired using a computing system equipped with an Intel Core i7-8700K CPU, 4.70-GHz, and CPU 62-GB RAM.

A. Datasets

The first hyperspectral dataset is the Airport dataset, whose spatial size is 100×100. Airport dataset was manually obtained by the AVIRIS over Los Angeles and Gulfport, USA. Features like roads and buildings on the Earth's surface are considered as background, while anomalies refer to flights of different heights. Fig. 4(a) and (b) gives the false-color images and the ground-truth maps.

The second hyperspectral dataset, the Urban dataset, comprises five HSI of spatial size 100×100 extracted from AVIRIS data. Anomalies within these images are identified as roofs and other objects that exhibit substantial deviations from their surrounding environment. We present the false-color images and the ground-truth maps in Fig. 5(a) and (b).

The final experimental data were obtained from the SpecTIR Hyperspectral Airborne Rochester Experiment (SHARE), featuring a spectral resolution of 5 nm and a ground spatial resolution of 1 m. For evaluation purposes, we have chosen an image area measuring 180×180 pixels with 120 bands. Within this image scene, the predominant normal classes, including roads and vegetation, constitute the background. Conversely, anomalies consist of different size man-made colorful square fabrics. Fig. 6(a) and (b) displays the false-color image and the ground-truth map.

B. Detection Performance

The proposed method is compared with several prominent AD approaches, including the classic RX [2], three LRaSMD models (LSMAD [18], anomaly detection method based on low-rank and sparse representation (LRASR) [19], and anomaly detection method based on graph and TV regularized LRR (GTVLRR) [22]), three tensor representation models (TPCA [33], PCA-TLRSR [3], and PTA [36]) and two deep learning-based AD methods (LREN [41] and DeCNN-AD [42]). For parameter selection, we optimize parameters based on their articles to attain first-class detection accuracy. Specifically, the parameter rank in LSMAD varies from 1 to 10. As for LARSR, two parameters β and λ are, respectively, selected from $\{0.001, 0.01, 0.05, 0.1, 0.5, 1\}$. In terms of GTVLRR, three regularization parameters $\beta=0.2$, $\gamma=0.05$ and λ varies from 0.1 to 1. Three hyper-parameters in PCA-TLRSR and four hyper-parameters in PTA are determined according to [3] and [36], respectively. For LREN, we set the hidden nodes= 9 or 10, k selected from $\{5, 6, 7, 8, 9\}$, and λ selected from $\{0.001, 0.01, 0.1, 1\}$. As for DeCNN-AD, three parameters K varies from 1 to 10, the

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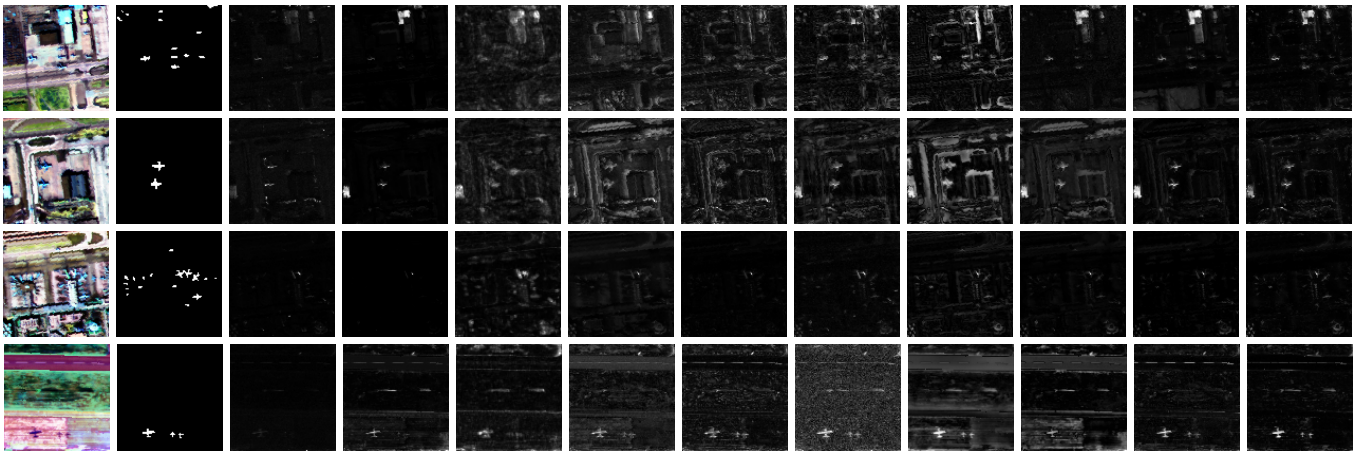


Fig. 4. (a) False-color HSIs, (b) ground truth, detection maps of (c) RX, (d) LSMAD, (e) TPCA, (f) LRASR, (g) GTVLRR, (h) PCA-TLRSR, (i) PTA, (j) LREN, (k) DeCNN-AD, and (l) TDNN-TA on the Airport datasets. (Top to bottom) Images are Airport 1 to 4.

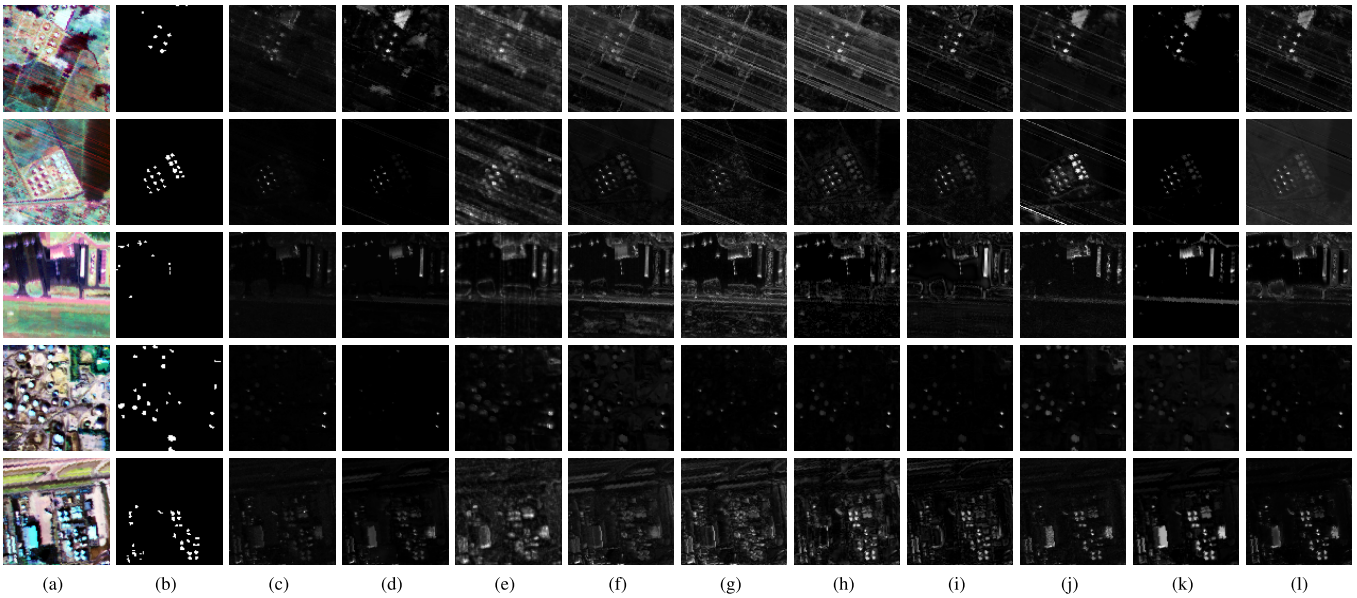


Fig. 5. (a) False-color HSIs, (b) ground truth, detection maps of (c) RX, (d) LSMAD, (e) TPCA, (f) LRASR, (g) GTVLRR, (h) PCA-TLRSR, (i) PTA, (j) LREN, (k) DeCNN-AD, and (l) TDNN-TA on the Urban datasets. (Top to bottom) Images are Urban 1 to 5.

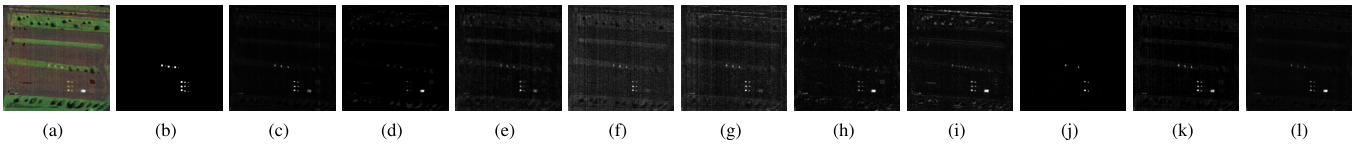


Fig. 6. (a) False-color HSIs, (b) ground truth, detection maps of (c) RX, (d) LSMAD, (e) TPCA, (f) LRASR, (g) GTVLRR, (h) PCA-TLRSR, (i) PTA, (j) LREN, (k) DeCNN-AD, and (l) TDNN-TA on the SpecTIR dataset.

values of β and λ change from 10^{-4} to 1. The parameters in our model are selected to the suitable values, which will be discussed in Section V-C.

To comprehensively evaluate the detection performance, we conduct both qualitative and quantitative analyses. In addition to visually inspecting the resultant anomaly maps, we utilize three evaluation indicators to quantitatively evaluate the detection accuracy across a range of compared methods, including the receiver operating characteristic (ROC) curve [51], the area under the ROC curve (AUC) [52] and the anomaly background separation map. The ROC curve illustrates the dynamic correlation between the probability of detection (P_d) and the false alarm rate (P_f) for a broad spectrum of possible thresholds, with superior models exhibiting higher P_d values for a given P_f. The AUC, calculated as the area under this curve, represents an ideal value of 1. The anomaly background separation map visually illustrates the degree of separation between the background and anomaly components through the box plot representation of each component. A wider gap between the background box

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TABLE III
AUC VALUES ON ALL TESTING DATASETS

HSI Dataset	RX	LSMAD	TPCA	LRASR	GTVLRR	PCA-TLRSR	PTA	LREN	DeCNN-AD	TDNN-TA
Airport-1	0.8221	0.8133	0.8023	0.8020	0.9146	0.9380	0.9148	0.8423	0.8774	**0.9433**
Airport-2	0.8404	0.9704	0.8891	0.8843	0.8872	0.9744	0.9379	0.9384	0.9644	**0.9787**
Airport-3	0.9288	0.9432	0.9297	0.9413	0.9225	0.9552	0.9422	0.9307	0.9517	**0.9576**
Airport-4	0.9526	0.9414	0.9432	0.9813	0.9890	0.9940	0.9976	0.8376	0.9926	**0.9982**
Urban-1	0.9907	0.9620	0.9390	0.9700	0.9235	0.9729	**0.9910**	0.9856	0.9886	0.9816
Urban-2	0.9946	0.9857	0.9409	0.9971	0.9079	0.9905	0.9968	0.9896	**0.9992**	0.9973
Urban-3	0.9513	0.9626	0.8224	0.9472	0.9564	**0.9833**	0.9709	0.9289	0.9380	0.9496
Urban-4	0.9887	0.9911	0.9617	0.9966	0.9653	0.9950	0.9965	0.9891	0.9959	**0.9969**
Urban-5	0.9692	0.9425	0.9370	0.9212	0.9411	0.9826	**0.9839**	0.9559	0.9629	0.9802
SpecTIR	0.9914	0.9552	0.9693	0.9553	0.9779	0.9805	0.9814	0.9840	0.9951	**0.9961**

and the anomaly box indicates better separation and detection performance. With these evaluation indicators, the detection performance can be summarized as follows.

Fig. 4 illustrates the detection maps of Airport datasets generated by various competing methods. The first two methods exhibit noticeable instances of missed detections. TPCA is poor in maintaining the shape of anomalies. It is apparent that LRASR, GTVLRR, PCA-TLRSR, and PTA display prominent false alarms, characterized by the presence of numerous regions of white suspended substances. These regions correspond to background pixels that have been erroneously classified as anomalies. While LREN effectively detects most of the anomaly pixels, it suffers from relatively poor background suppression performance. This phenomenon is more pronounced in Airport-2 and Airport-4. The same issue occurs with DeCNN-AD. While it has already performed well, there are still false alarms in the bottom left corner of Airport-1. Compared to other competing methods, the proposed TDNN-TA demonstrates a promising capability to detect anomalies and maintain a good balance between highlighting the anomaly and suppressing the background, with its detection map closely resembling the ground truth. Figs. 5 and 6 show the detection maps of Urban datasets and SpecTIR, respectively. In these figures, similar patterns are observed. Despite TDNN-TA exhibiting suboptimal performance on Urban-3, its overall detection results remain satisfactory. As for the SpecTIR dataset, the detection efficacy of DeCNN-AD and TDNN-TA is notably effective, surpassing that of other comparative methods. Upon closer inspection of the background in the upper left and lower right corners, it becomes apparent that while DeCNN-AD incorrectly classifies some background pixels as anomalies, TDNN-TA successfully avoids such misclassification.

Fig. 7 displays the ROC curves of the detection results corresponding to the competing methods on the ten datasets. It is obvious that the ROC curve of the proposed TDNN-TA takes the leading position on almost all datasets, indicating outstandingly effective detection performance. This observation aligns with the findings presented in Table III. Table III displays the AUC scores for various competing methods on all datasets, providing a quantitative evaluation. The table highlights the best results in boldface and the second-best results with underlining. It is evident that TDNN-TA achieves the best AUC value on all datasets, except for Urban datasets, indicating superior performance in terms of detection effectiveness. Even on the Urban dataset, one optimal and one suboptimal result were achieved by TDNN-TA.

Furthermore, as depicted in Fig. 8, the anomalies-background separability diagrams are employed to further validate the detection reliability of the compared methods. It is evident that the gap between the background box and the anomalies box is substantial, indicating a promising separability between anomalies and background by TDNN-TA. In addition, the consistently smaller size of the background boxes suggests a better ability of TDNN-TA to suppress background interference.

C. Parameter Analysis

This section specifically analyzes how the performance of the proposed TDNN-TA is influenced by key parameters. There are six variables supposed to be optimized to gain satisfactory detection results, including $\alpha = (\alpha_1, \alpha_2, \alpha_3)$, $\beta = (\beta_1, \beta_2, \beta_3)$, λ, t, patch size s_p and patch number n_p. We utilize the Airport datasets as an example to elucidate the parameter selection process and AUC value as the detection quality evaluation.

1) Impact of α and β: α regulates the proportion of each mode correlation of HSI. β controls the weighting of each dimension within each group. In order to reduce the number of parameters, we set $\alpha = (1, 1, \theta_1)/(2 + \theta_1)$ and $\beta = (1, 1, \theta_2)/(2 + \theta_2)$, where θ_1 and θ_2 are the balance parameter to control α and β. Fig. 9 presents the variation of AUC values with respect to θ_1 and θ_2. We can find the appropriate θ_1 easily from Fig. 9(a). When $\theta_1 \geqslant 0.7$, the AUC value reaches its optimum and approaches stability, which means the spectral dimension and spatial dimension have a similar contribution ratio, with the spatial dimension contributing slightly more, in the global low-rank constraint. From Fig. 9(b), it can be observed that θ_2 could be set around 0.8, which implies that after grouping, the intragroup low-rank structure makes a

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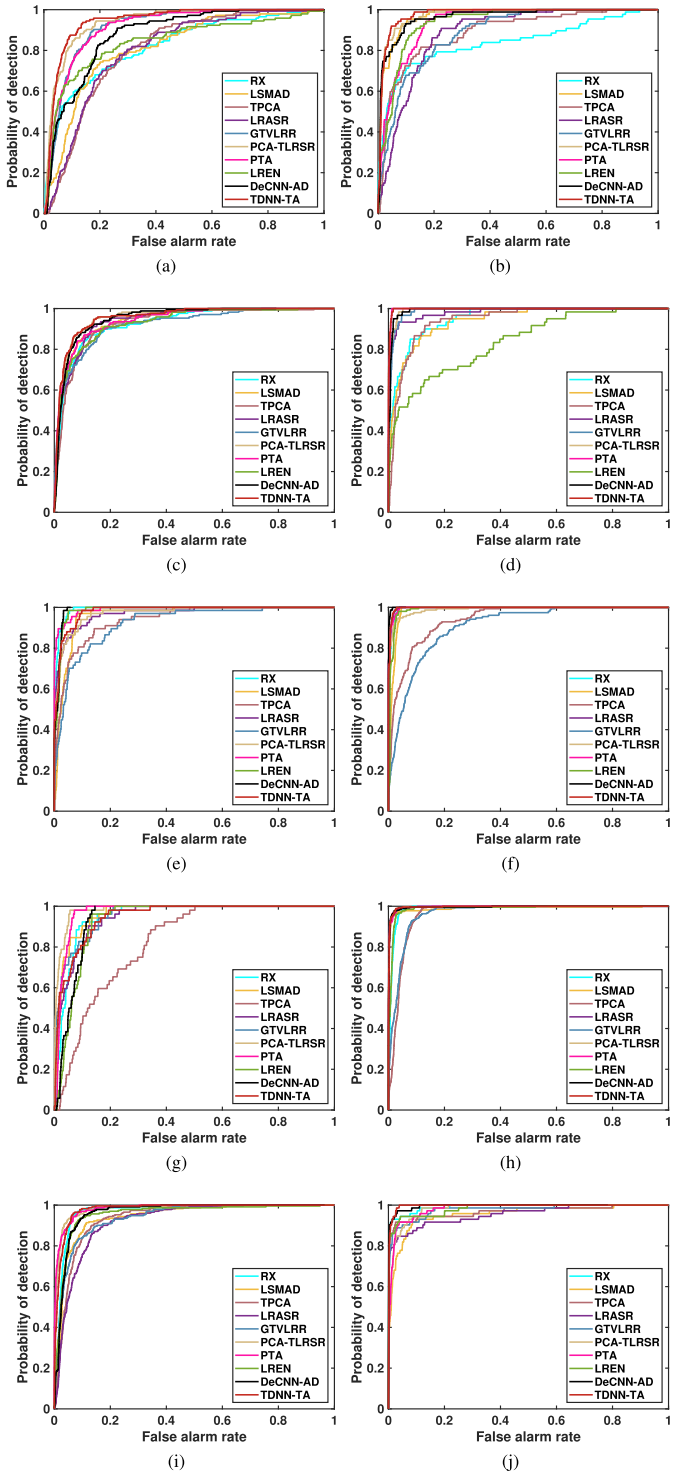


Fig. 7. ROC curves of (P_d, P_f) on (a) Airport-1, (b) Airport-2, (c) Airport-3, (d) Airport-4, (e) Urban-1, (f) Urban-2, (g) Urban-3, (h) Urban-4, (i) Urban-5, and (j) SpecTIR.

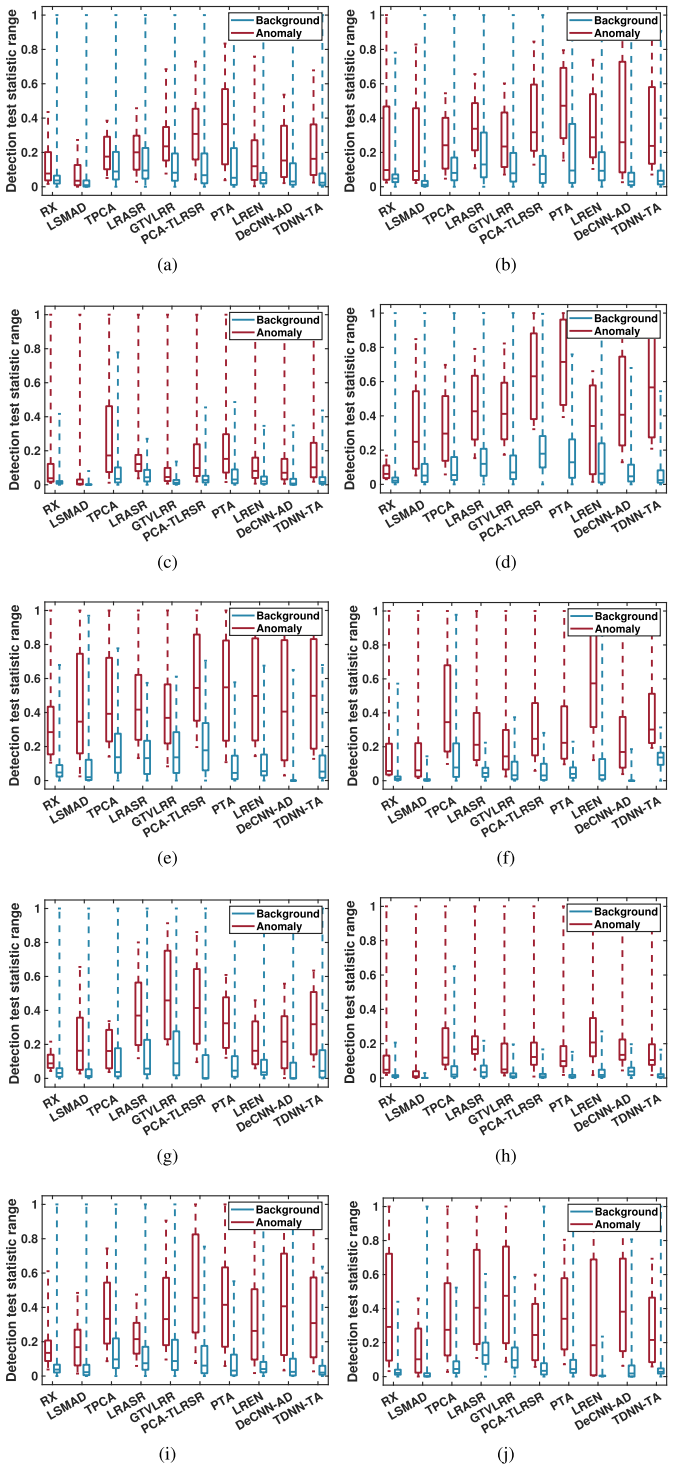


Fig. 8. Separability diagrams of the competing methods on (a) Airport-1, (b) Airport-2, (c) Airport-3, (d) Airport-4, (e) Urban-1, (f) Urban-2, (g) Urban-3, (h) Urban-4, (i) Urban-5, and (j) SpecTIR.

significant contribution, allowing for the full exploitation of nonlocal self-similarity. In other words, our model effectively leverages the nonlocal self-similarity of HSI.

2) Impact of λ and t: Fig. 10 demonstrates the regularization parameters λ and t. When $t > 0.3$, the AUC value in our method is nearly stable. However, it is sensitive to λ. It especially achieves the highest AUC value when $\lambda = 0.04$.

3) Impact of s_p and n_p: Fig. 11 illustrates the impact of patch size on our model. It can be observed that our model is robust to the number of patches within a group, while it is sensitive to the patch size. The specific patch size is closely related to the intrinsic characteristics of HSI. However, we observe that setting the patch size to 5×5 achieves good results across multiple datasets.

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TABLE IV
AUC VALUES OF ALL COMPARISON METHODS UNDER DIFFERENT NOISE LEVELS

Noise Level	RX	LSMAD	TPCA	LRASR	GTVLRR	PCA-TLRSR	PTA	LREN	DeCNN-AD	TDNN-TA
0.05	0.9743	0.9544	0.9758	0.9515	0.9622	0.9756	0.9769	0.9404	0.9808	**0.9898**
0.1	0.9318	0.9488	0.9681	0.9409	0.9479	0.9603	0.9677	0.9293	0.9518	**0.9803**
0.15	0.8697	0.9431	0.9432	0.9100	0.9254	0.9386	0.9475	0.8934	0.8996	**0.9526**
0.2	0.8241	0.9435	0.9137	0.9011	0.7674	0.9105	0.9451	0.8920	0.8205	**0.9578**
0.25	0.8244	0.9291	0.9005	0.8736	0.5925	0.8756	0.9042	0.8974	0.7390	**0.9309**
0.3	0.7857	0.9215	0.8628	0.8405	0.5753	0.8549	0.8922	0.8715	0.7137	**0.9248**

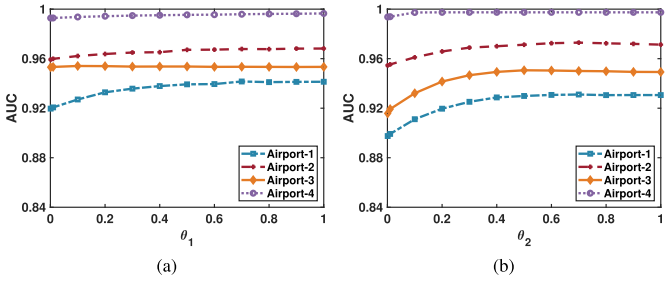


Fig. 9. (a) θ_1 and (b) θ_2 tuning on Airport datasets.

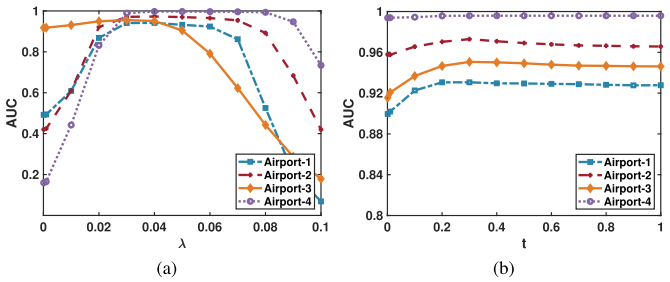


Fig. 10. (a) λ and (b) t tuning on Airport datasets.

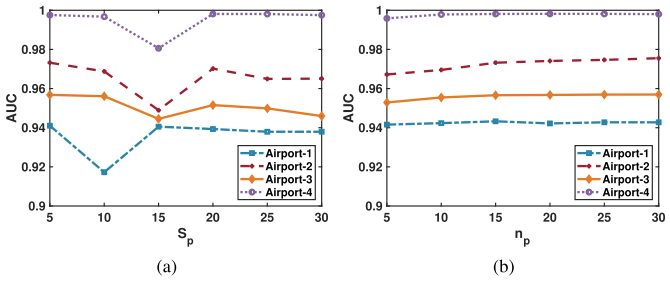


Fig. 11. (a) Patch size and (b) patch number tuning on Airport datasets.

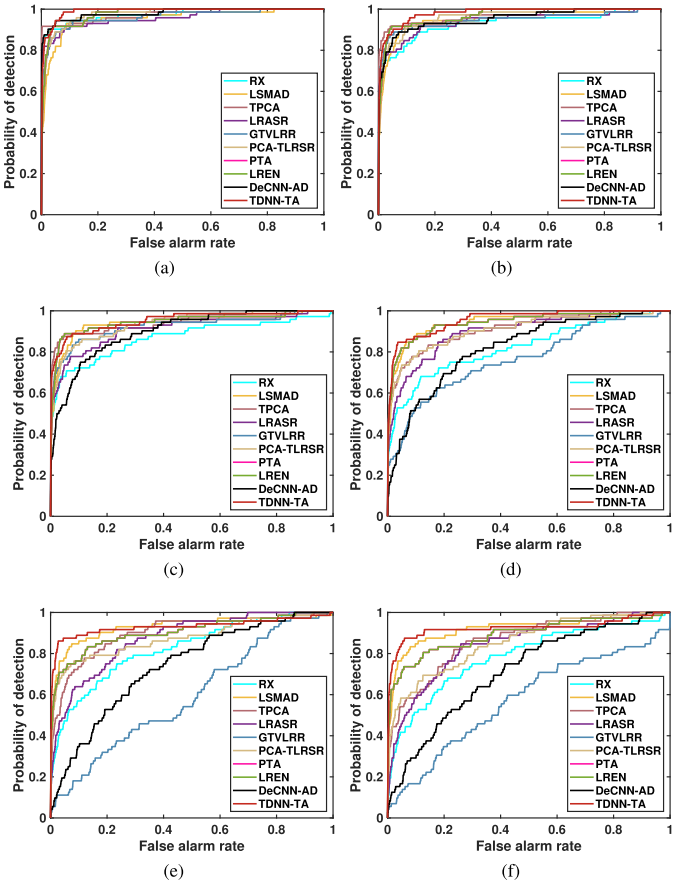


Fig. 12. ROC curves of (P_d, P_f) on SpecTIR by ten different methods under different Gaussian noise. (a) Variances = 0.05. (b) Variances = 0.1. (c) Variances = 0.15. (d) Variances = 0.2. (e) Variances = 0.25. (f) Variances = 0.3.

D. Discussion

1) Detection Performance under Different Noise: To thoroughly investigate the performance of the proposed TDNN-TA under different noise, we added Gaussian noise with different levels of variance to the SpecTIR dataset and compared the results with other methods. The added Gaussian noise has a mean of 0, with variances of 0.05, 0.1, 0.15, 0.2, 0.25 and 0.3. The AUC values of all comparison methods under different noise levels are presented in Table IV. As shown in Table IV, TDNN-TA maintains optimal detection performance under varying noise intensities, indicating that the proposed method exhibits a certain level of robustness to noise conditions. Fig. 12 displays the ROC curves of the detection results corresponding to the competing methods under different noise. It is evident that the ROC curve of the proposed TDNN-TA outperforms other methods, demonstrating its exceptional detection performance. This observation is consistent with the results presented in Table IV.

2) Detection Performance With Different Size Targets: To further explore the detection performance of the proposed TDNN-TA on targets of different sizes, we designed the following experiment. In this experiment, a simulated dataset is generated based on the real hyperspectral imagery, which was collected by the airborne visible/infrared imaging spectrometer

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TABLE V
TIME CONSUMPTION (S) OF COMPETING METHODS ON ALL DATASETS

HSI Dataset	Airport-1	Airport-2	Airport-3	Airport-4	Urban-1	Urban-2	Urban-3	Urban-4	Urban-5	SpecTIR
RX	0.14	0.11	0.51	0.55	0.33	0.31	0.28	0.28	0.41	0.82
LSMAD	11.85	10.56	11.44	10.22	11.22	11.14	10.53	10.95	11.63	19.96
TPCA	17.47	15.97	16.03	15.54	15.54	16.04	15.29	16.38	17.30	27.84
LRASR	54.93	39.64	62.31	61.08	48.81	23.28	37.72	49.00	64.14	107.58
GTVLRR	231.02	212.17	183.93	194.86	224.00	195.90	205.12	157.68	209.28	544.38
PCA-TLRSR	2.31	1.59	9.54	10.09	7.79	0.92	2.22	0.89	0.87	7.17
PTA	48.23	45.96	46.60	43.23	41.93	59.38	38.36	36.34	41.76	63.22
LREN	356.03	357.51	362.63	213.96	335.43	346.82	319.28	351.77	343.30	975.39
DeCNN-AD	27.10	19.65	15.64	39.11	7.29	7.30	7.13	7.45	7.37	38.70
TDNN-TA	1041.59	1869.88	1343.13	1144.23	1054.26	745.75	1600.26	1631.92	1941.01	1482.53

over San Diego airport in California, USA. After removing some degraded spectral bands, there are 189 spectral bands left. We select a 100 × 100 region (pixels in rows 71–170 and columns 41–140) from the original image to construct the simulated data. The target spectrum is t selected from the pixel spectrum at (22, 70) within the same dataset (which is inherently an anomalous spectrum in the original image). The targets are embedded according to the following replacement signal model:

$$\mathbf{y} = (1-\alpha)\mathbf{b} + \alpha\mathbf{t}, \quad 0 < \alpha \leq 1 \tag{24}$$

where $\mathbf{b}$ is the original spectrum, $\mathbf{t}$ is the target spectrum, $\mathbf{y}$ is the filled spectrum, and α is the fill coefficient.

In the first simulated data, nine target blocks distributed in three rows and three columns have been embedded. The sizes of the targets in each row are 3 × 3, 5 × 5, and 7 × 7 from top to bottom, respectively. The false-color image and ground-truth map of the first simulated data are shown in Fig. 13(a) and (b), respectively. In the second simulated data, nine target blocks, which have the same size are embedded at nine randomly selected locations. The false-color image and ground-truth map of the second simulated data are shown in Fig. 13(d) and (e), respectively. The fill coefficient is randomly selected from the range of 0.8 to 1.

The AUC scores for our method on two datasets are 0.9914 and 0.9922, respectively. The detection results using are shown in Fig. 13(c) and (f). As can be seen from the detection maps, our method achieves excellent detection performance for targets of large, medium, and small sizes, particularly for small targets.

3) Convergence Analysis: Fig. 14 is the numerical convergence results on all datasets, in which all other parameters are guaranteed to be optimal. From Fig. 14(a), we can observe that, after iteration over 30, the AUC value becomes stable by degrees. In real scenarios, AUC is often not computable. Therefore, we also analyze the iteration error of anomaly tensor $\mathcal{S}$, i.e., $\|\mathcal{S}^{k-1} - \mathcal{S}^{k}\|_F$ (for the kth iteration) with respect toss the number of iterations as shown in Fig. 14(b). From the figure, it can be observed that the convergence difference gradually decreases and approaches 0 after 30 iterations. As a result, our algorithm is numerically convergent.

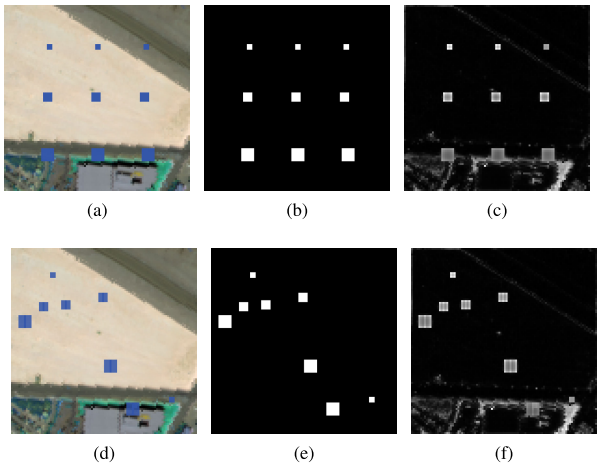


Fig. 13. (a) False-color image, (b) ground-truth map, and (c) detection map by TDNN-TA of the first simulated data. (d) False-color image, (e) ground-truth map, and (f) detection map by TDNN-TA of the second simulated data.

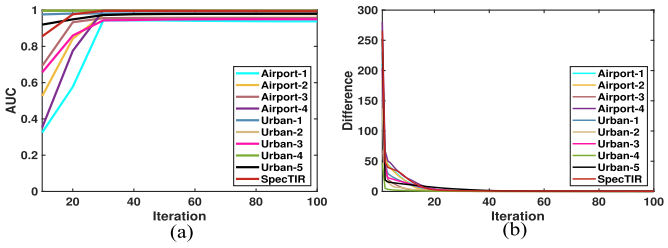


Fig. 14. Change of (a) AUC value and (b) iteration error of $\mathcal{S}$ with iterations on all datasets.

4) Running Time and Complexity: Finally, the running time of the competing methods on all datasets is summarized in Table V. Notably, RX, LSMAD, and TPCA demonstrate the most satisfactory computation speed, attributed to their simple and efficient algorithms. LRASR, PCA-TLRSR, and PTA also demonstrate remarkable performance. The results of DeCNN-AD do not include training time. The proposed model prioritizes enhancing accuracy. Due to the redundancy of the framelet transform, the processing time will inevitably be prolonged. In Algorithm 1, the most computationally expensive part is updating $\mathcal{L}$ and $\mathcal{Z}_k$. The time for computing subproblem $\mathcal{L}$ is mainly consumed on FFT and t-SVD, and

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the complexity is $\mathcal{O}(N \cdot (n_p \min(s_p^4 n_3,\, s_p^2 n_3^2) + s_p^2 \min(n_3^2 n_p,\, n_p^2 s_p^2) + n_3 \min(s_p^4 n_p,\, s_p^2 n_p^2) + s_p^2 n_3 n_p(\log(s_p^2) + \log(n_3) + \log(n_p))))$; the complexity of computing subproblem $\mathcal{Z}_k$ is $\mathcal{O}(wn_1 n_2 n_3 + \min(n_1^2 n_2,\, n_2^2 n_1) wn_3 + \min(n_1^2 n_3,\, n_3^2 n_1) wn_2 + \min(n_3^2 n_2,\, n_2^2 n_3) wn_1)$. The computational complexity of the other parts is $\mathcal{O}(n_1 n_2 n_3)$. However, there is room for improvement in terms of computational time. Parallel computation for the framelet transform along different directions and minimizing calculations in each group can significantly reduce the computational complexity of this algorithm. This optimization is a promising direction for future research. This improvement should also be considered for future work.

VI. Conclusion

In this article, we propose a TDNN-based tensor approximation model for HAD, and develop an efficient ADMM-based algorithm with empirical convergence analysis. We leveraged the F-3MTNN to the group tensors and the 3DTNN to the background tensor. The redundancy of the framelet allows us to precisely the high correlation in the spatial and spectral dimensions, as well as the nonlocal self-similarity. The proposed TDNN takes advantage of the inherent properties of different transforms, enabling a flexible and accurate exploration of the background prior, including more spatial similarities and nonlocal self-similarity. We conducted experiments on three datasets, covering a total of ten different data instances, and the proposed method consistently achieved promising results, outperforming others on six of these datasets. In addition, we explored the performance under varying noise intensities and different target sizes. Despite different levels of noise, our method still achieved the best detection performance, which means that the method is robust. For targets of varying sizes, our method is relatively more suited for smaller-sized targets. Further improvements in this area will be a focus of future work.

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